

PHANTOM DETECTIVE

Game Design & Project Documentation

Unity Horror Game | First-Person Survival

Engine	Unity (URP)
Genre	Horror / Survival / Puzzle
Perspective	First-Person
Players	Single Player
Levels	1 Level — Abandoned Asylum
Setting	24 Rooms across 2 Floors + Basement

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1. Project Description

What does the player do?

In Phantom Detective, the player takes on the role of a renowned detective hired by wealthy patrons to investigate an abandoned asylum. Armed with only a flashlight, the player must explore 24 rooms spread across two floors and a basement, collecting scattered memory fragments — pieces of a jigsaw puzzle hidden throughout the environment. The player navigates the asylum carefully, opens doors with a click of the mouse, crouches to stay hidden from roaming ghosts, and pieces together the puzzle to reveal the truth behind each spirit's death.

Why do they do it?

Each of the three ghosts haunting the asylum is a trapped soul, endlessly reliving the loop of its own murder. The player's mission is to uncover the dark secret buried within the asylum — to discover how each victim was killed and reconstruct the crime. Completing the jigsaw puzzle for each ghost is the only way to free their spirit and ultimately escape the asylum. The player is driven not just by survival, but by the compulsion to solve the mystery and bring peace to those who cannot rest.

Where does the player do it?

The game takes place entirely within a massive, crumbling Abandoned Asylum — a sprawling structure of 24 rooms distributed across two floors and a basement. Every room is a potential crime scene: darkened corridors, rotting furniture, flickering shadows, and an ever-present sense that the walls themselves remember the horrors that occurred within them. The environment is drenched in atmospheric fog, oppressive darkness, and environmental storytelling, ensuring that every step deeper into the asylum feels heavier than the last.

How does the player do it?

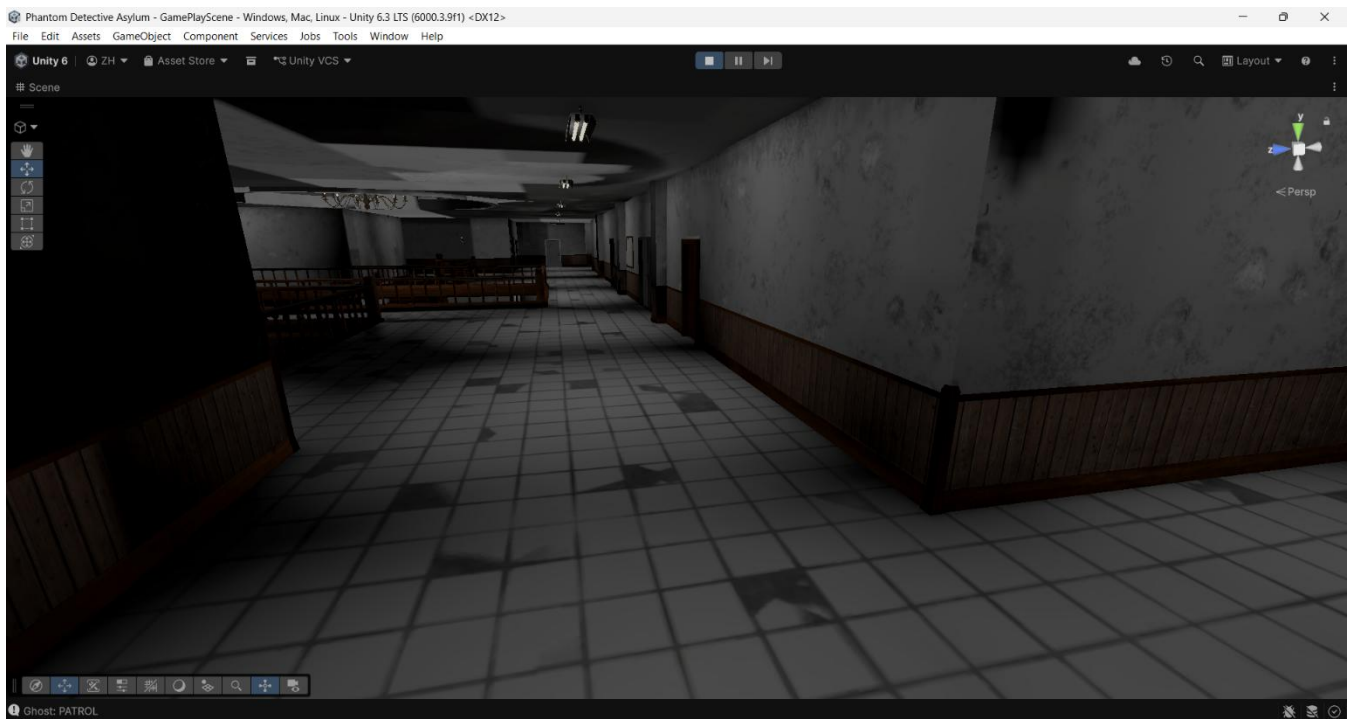
The player's sole tool is a flashlight — used to illuminate the darkness, reveal interactive objects, and highlight collectible memory fragments. Doors are opened with a mouse click. The player can crouch (Ctrl or C) to reduce sound and visibility, making it possible to slip past the three ghosts patrolling the asylum. Memory fragments are scattered throughout the rooms and must all be collected before they can be assembled on a puzzle board. Successfully completing the jigsaw reconstruction of each ghost's crime scene is the win condition of the game.

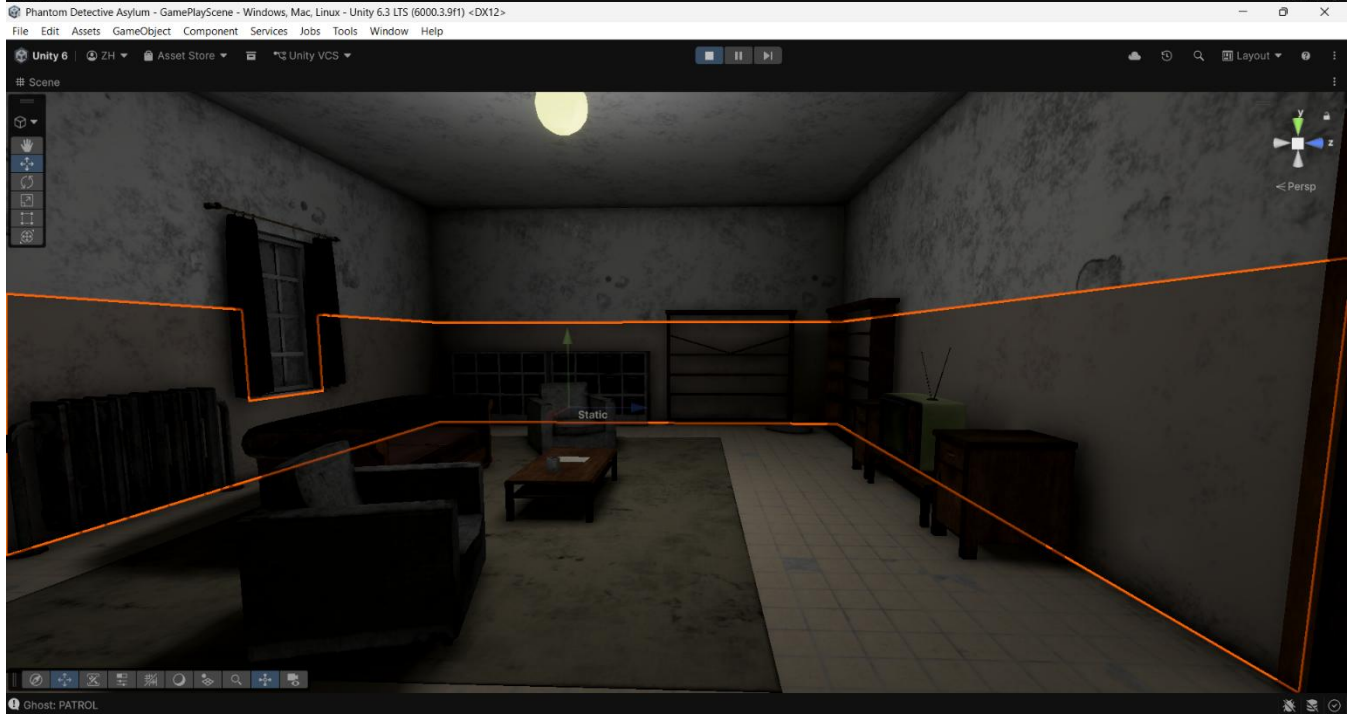
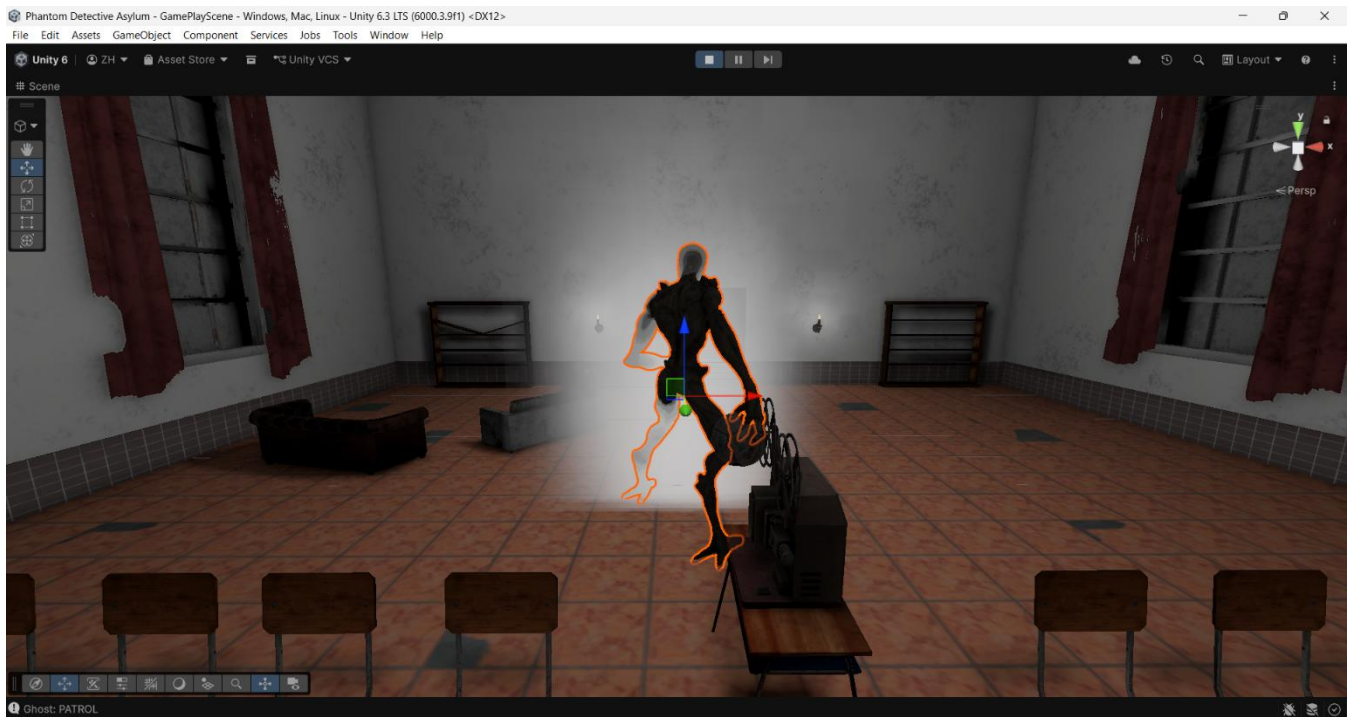
What emotion or experience should the player feel?

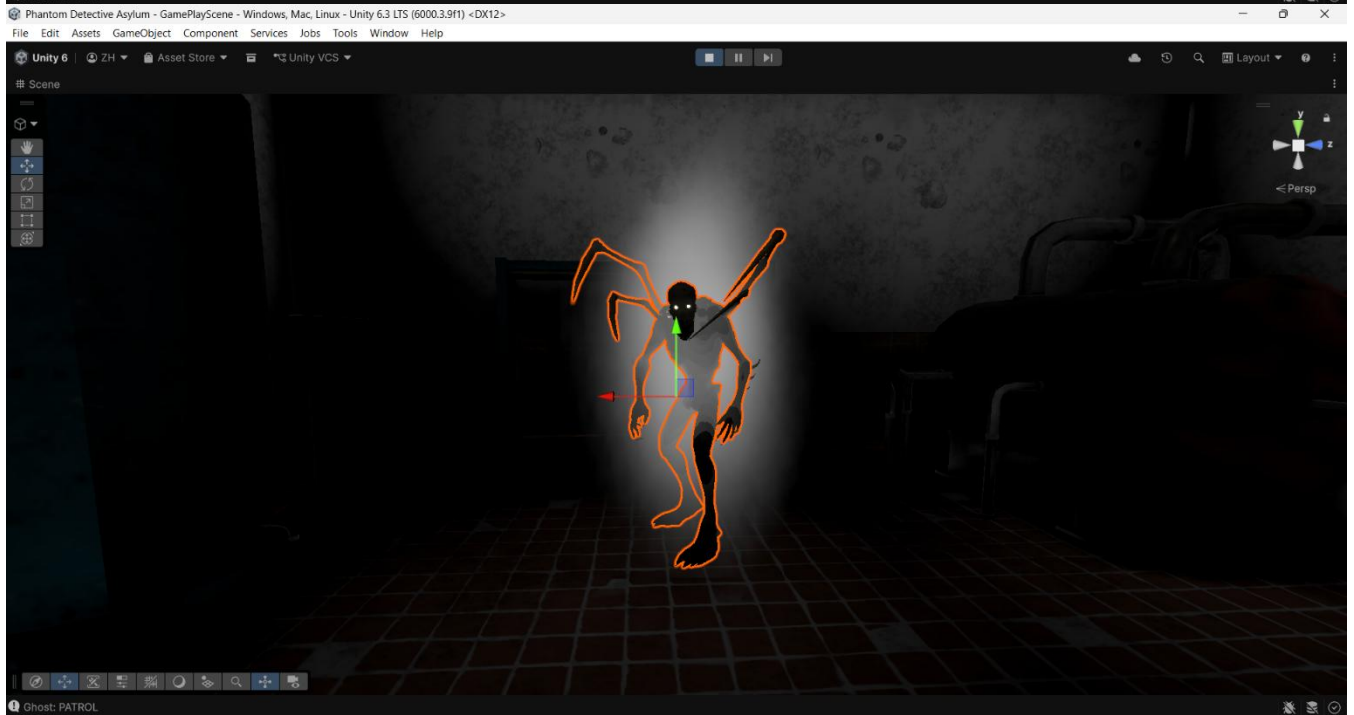
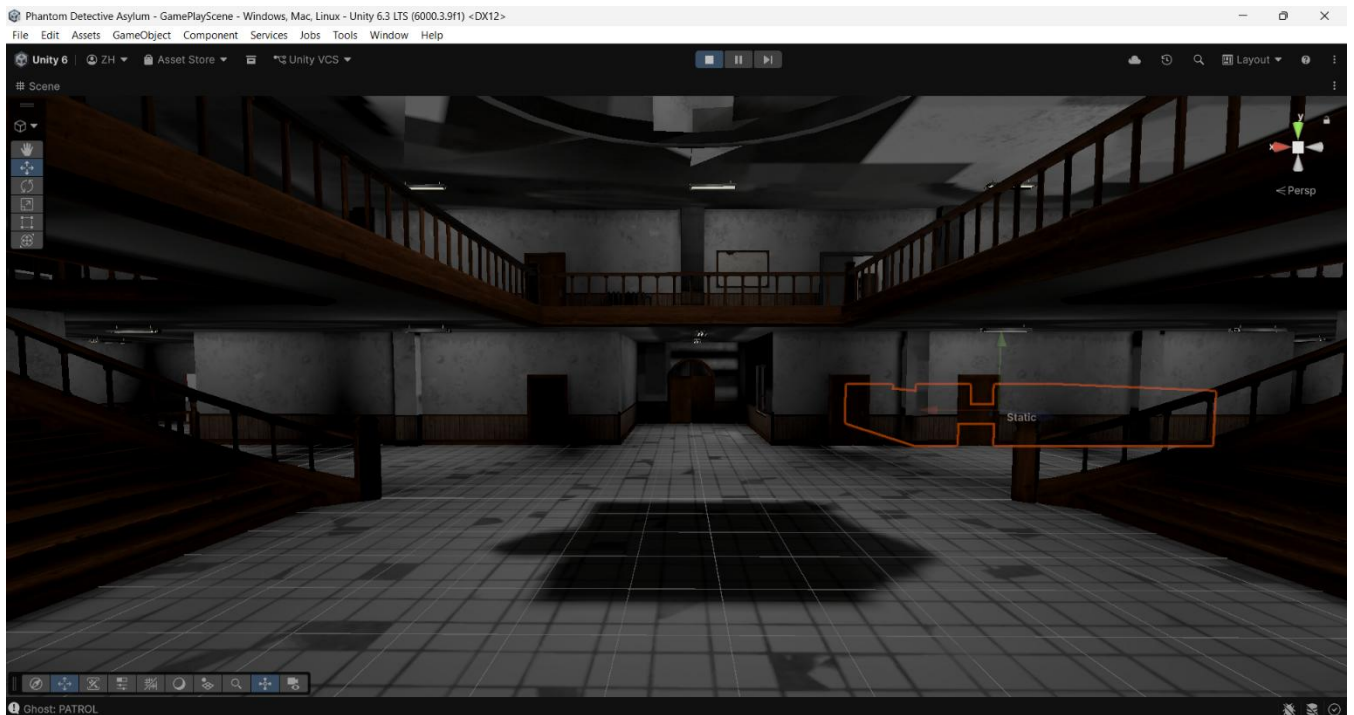
Phantom Detective is designed to evoke a persistent sense of melancholic curiosity — the painful pleasure of uncovering tragedy piece by piece. The player should feel the creeping dread of never being truly alone in the dark, the tension of breath held while a ghost passes inches away, and finally, the quiet emotional relief of freeing a lost soul. The atmosphere balances horror and sadness: this is not a game about jump scares, but about the slow revelation of a mystery in a place where the dead cannot move on.

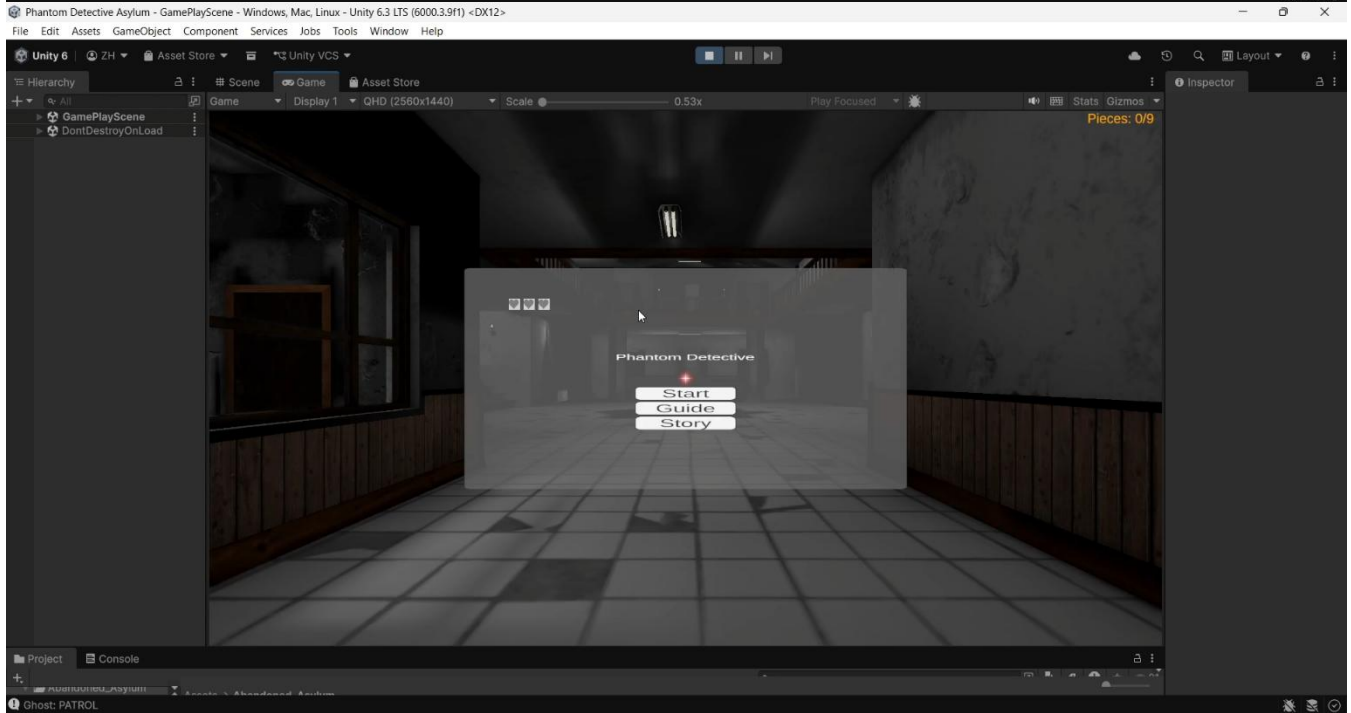
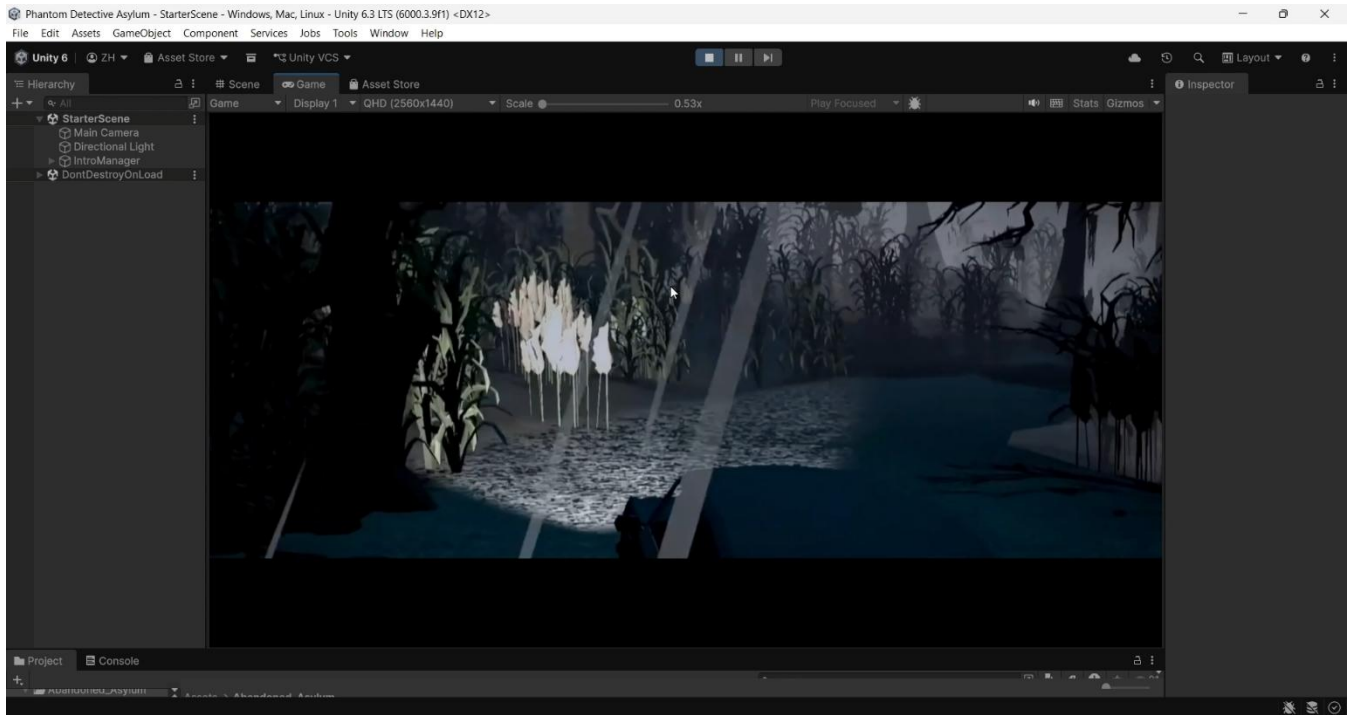
2. Screenshots

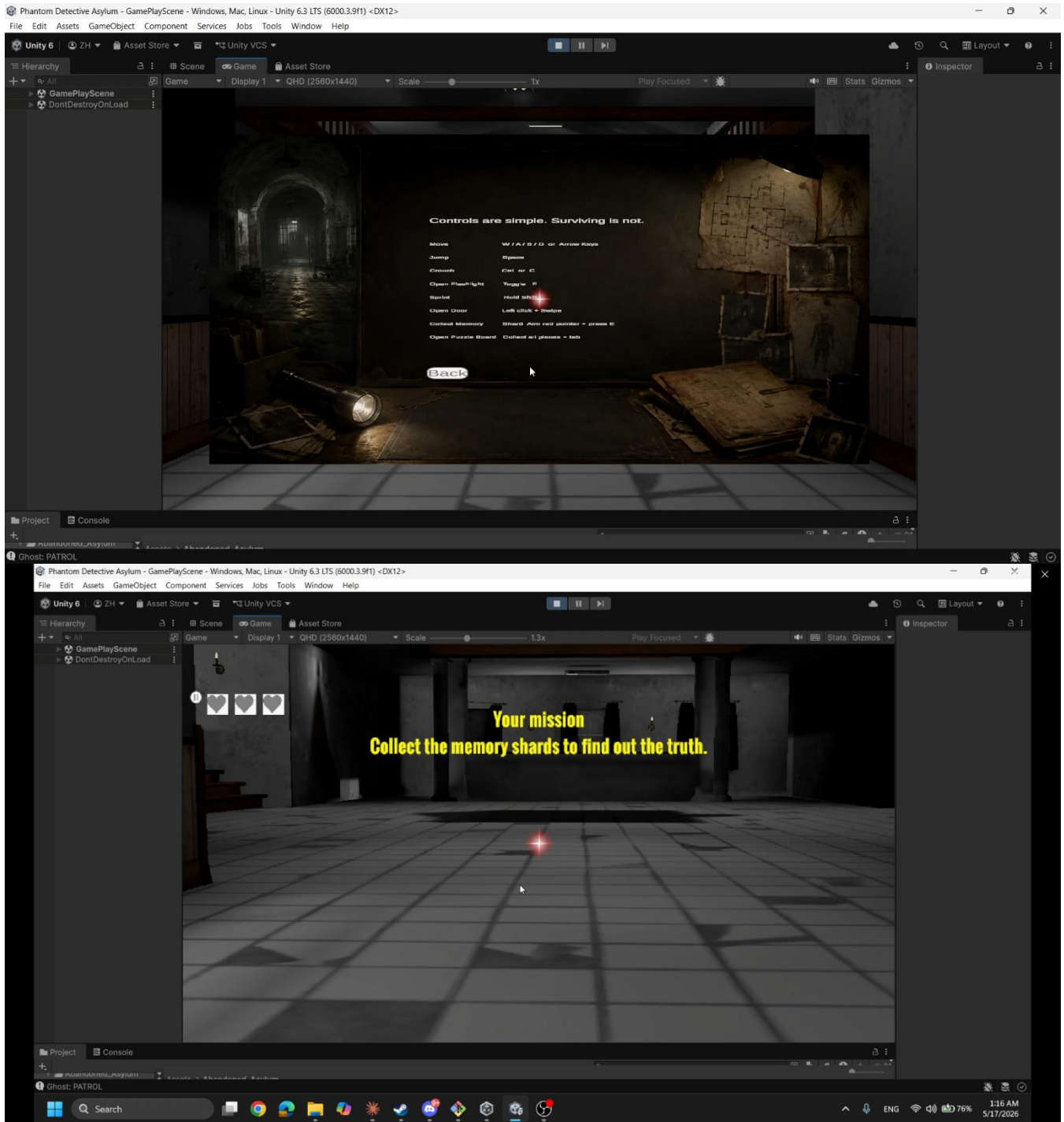
The following screenshots represent key visual moments from the game.

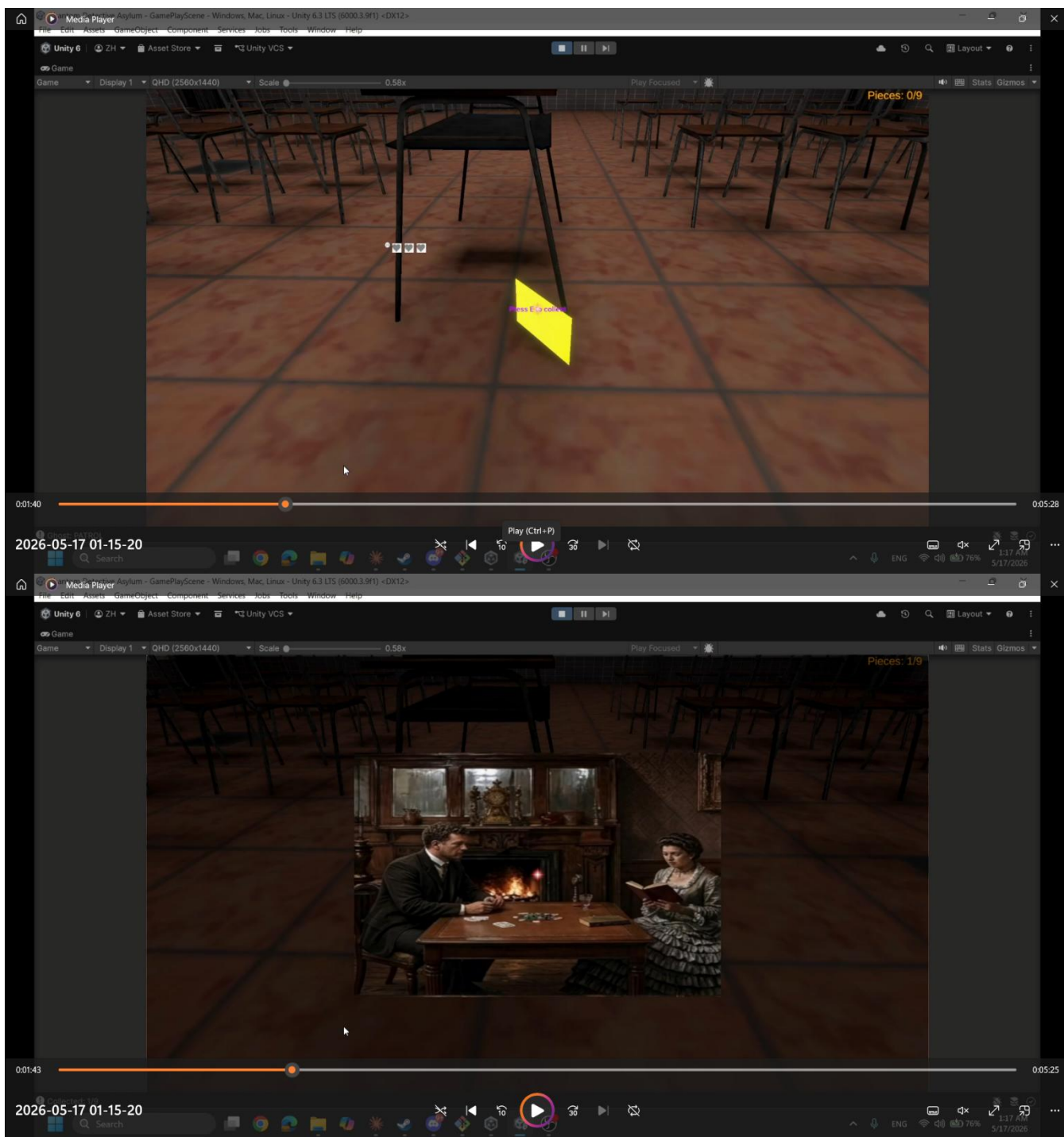






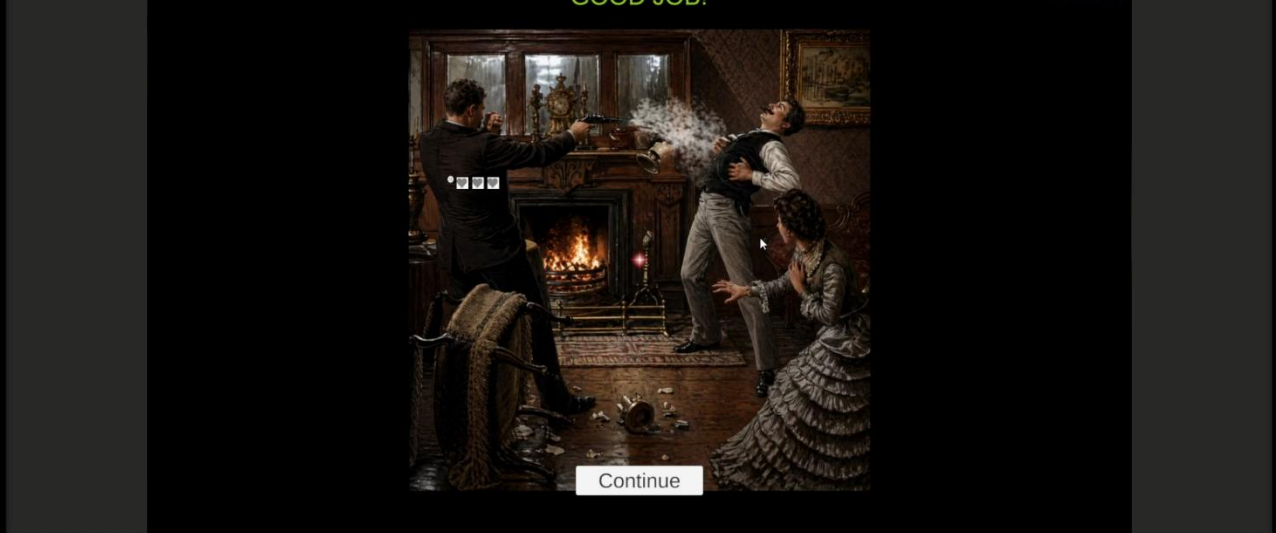




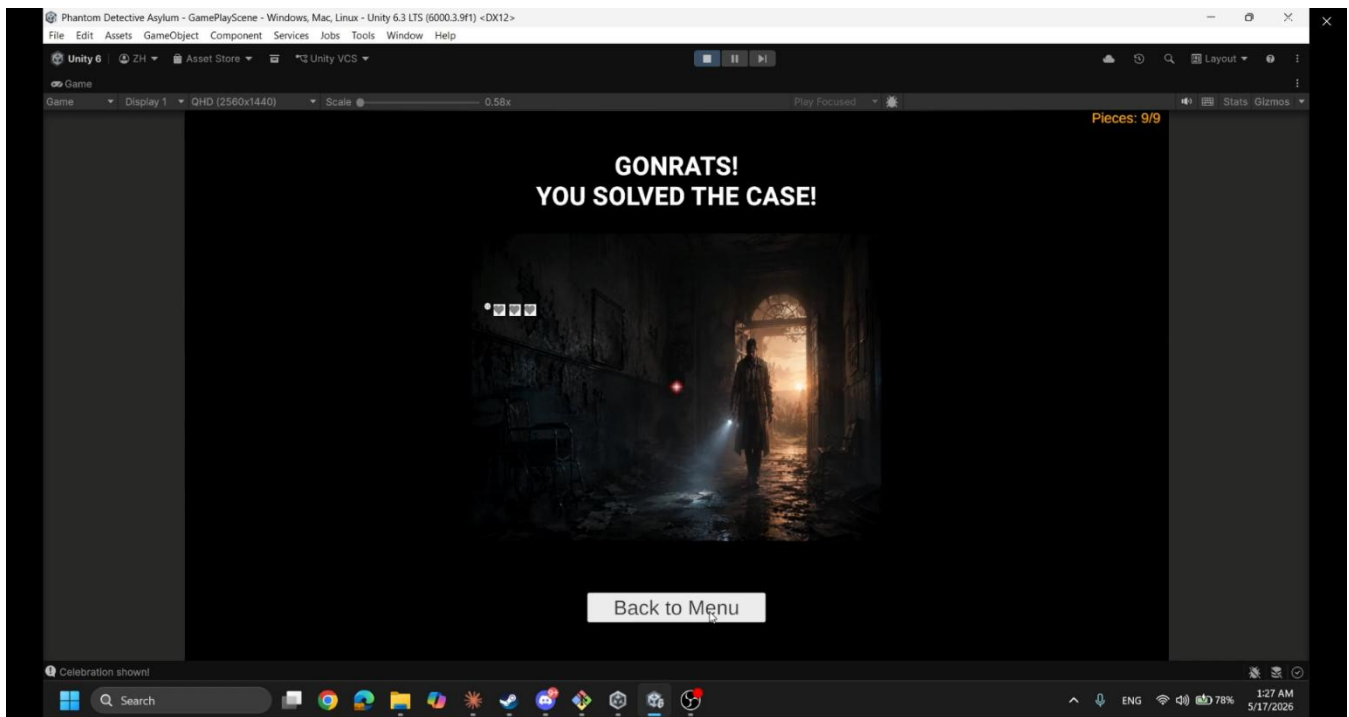




Player entered hiding spot — HIDDEN



PUZZLE COMPLETE!



3. User Manual

Getting Started

- Launch the game executable (PhantomDetective.exe)
- From the Main Menu, click Play to begin
- You will wake inside the abandoned asylum — your mission begins immediately

Controls

Action	Control	Description
Move	W / A / S / D	Walk through the asylum
Look Around	Mouse	Rotate camera view
Open / Close Doors	Left Mouse Click	Interact with doors
Crouch	Ctrl or C	Lower profile to hide from ghosts
Toggle Flashlight	F	Turn flashlight on/off
Pause Menu	Press button	Pause and access options
Collect Fragments	E	Approach and collect
Elevator	Y	Ascending and Descending

Gameplay Objectives

- Explore the abandoned asylum across 2 floors and a basement (24 rooms total)
- Locate and collect all memory fragments hidden throughout the asylum
- Avoid being detected by any of the 3 ghosts patrolling the building
- Once all fragments are collected, access the Puzzle Board
- Arrange the fragments correctly to reconstruct the crime scene image
- Successfully completing the puzzle reveals the truth — and you win

Survival Tips

- Use your flashlight sparingly — light can attract ghost attention
- Crouch (Ctrl / C) when a ghost is nearby to reduce your sound and visibility
- Ghosts patrol fixed loops — learn their routes before moving
- Each ghost has specific behaviours; observe carefully before approaching
- If a ghost is disturbed, the game becomes harder — avoid detection at all costs
- Check every room — fragments can be in unexpected locations

Winning the Game

Collect every memory fragment scattered across the asylum, then navigate to the Puzzle Board. Place all fragments in the correct positions to reconstruct the crime scene image. Once the puzzle is complete, a final event will trigger — you have uncovered the truth and freed the lost souls. Congratulations.

4. Innovation & Creativity

What Makes Phantom Detective Unique?

Phantom Detective stands apart from conventional horror games by combining first-person survival stealth with a narrative puzzle-reconstruction mechanic. Rather than defeating enemies or simply escaping, the player must piece together the story of each ghost's death — making the act of solving the mystery the core gameplay loop. The ghosts are not obstacles to be overcome; they are the protagonists of a tragedy the player is there to witness and resolve.

Inspirations

The game draws inspiration from titles such as *Amnesia: The Dark Descent* (environmental horror and helplessness), *Phasmophobia* (ghost behaviour and tension), and puzzle-driven narrative games. However, *Phantom Detective* introduces its own unique blend: the investigator's perspective, the

ancient-lens mechanic, and the emotional resolution of freeing trapped souls rather than simply surviving.

Creative Mechanics & Storytelling

- Jigsaw Puzzle as Narrative Tool: Each completed puzzle is not just a win condition — it is a crime scene reconstruction that reveals how the ghost was murdered, making puzzle-solving emotionally meaningful
- Ghost as Case File: Each of the 3 ghosts represents a separate case. Every room is a crime scene. The player is always a detective, not just a survivor
- Light as Dual Risk: The flashlight is the only tool but also a liability — it illuminates the path while potentially revealing the player to nearby ghosts
- No Traditional Game Over: Instead of a hard fail state, disturbing ghosts triggers cumulative difficulty increases — the game punishes carelessness without breaking immersion
- Atmosphere over Action: The game prioritises slow dread, emotional tension, and melancholic storytelling over jump scares or combat
- Abandoned Asylum Setting: 24 rooms across 2 floors and a basement create a large, interconnected world that rewards thorough exploration

5. Core Mechanics

The core mechanics of Phantom Detective define how the game operates at its most fundamental level — the rules, data, and algorithms that govern every interaction between the player, the environment, the ghosts, and the puzzle. They consist of three interlocking elements: Resources, Entities, and Mechanics (events, processes, and conditions), all of which feed the game's internal economy and generate meaningful gameplay.

5.1 Resources

Resources are the quantifiable materials or values the game produces, consumes, trades, and destroys. In Phantom Detective the following resources are tracked numerically at all times:

- Memory Fragments — The primary collectible resource. The total count begins at 0 and increases by 1 each time the player picks up a fragment. The game tracks both the number collected and the number remaining in the world. Fragments cannot be destroyed or lost once collected.
- Ghost Alert Level — A numeric value (0–100) per ghost, representing how aware that ghost is of the player. It rises when the player is nearby or loud, and decays when the player hides or moves away. If any ghost's alert level reaches 100, a difficulty penalty is applied.
- Puzzle Completion Count — A numeric counter (0 to total pieces) tracking how many puzzle fragments have been correctly placed on the puzzle board. Reaching the maximum value triggers the win condition.
- Player Noise Level — A numeric value (0–100) generated by player movement. Walking produces moderate noise; crouching reduces it significantly; running (if implemented) maximises it. This value feeds directly into ghost alert calculations.

5.2 Entities

Entities are specific instances of a resource or the state of a game-world element. Simple entities are defined by a single value; compound entities are defined by a set of attributes, some of which may themselves be compound. Phantom Detective uses the following core entities:

Player (Compound Entity)

- Position — Numeric (X, Y, Z coordinates in world space)
- Stance — Symbolic (Standing / Crouching)
- Noise Level — Numeric (0–100); derived from stance and movement speed
- Lens Active — Symbolic (On / Off); controls highlight rendering of collectibles
- Fragments Collected — Numeric (0 to total fragment count)
- Flashlight State — Symbolic (On / Off) [cosmetic, also affects ghost line-of-sight]

Ghost (Compound Entity — one instance per ghost)

- Ghost ID — Symbolic (Ghost_01, Ghost_02, Ghost_03)
- Patrol State — Symbolic (Patrolling / Alerted / Searching / Returning)
- Alert Level — Numeric (0–100)
- Detection Radius — Numeric (metres); fixed per ghost, can scale with difficulty
- Current Waypoint — Symbolic (index into the ghost's defined patrol route)

Memory Fragment (Compound Entity)

- Fragment ID — Symbolic (unique identifier per piece)
- State — Symbolic (In World / Collected / Placed on Board)
- World Position — Numeric (X, Y, Z); valid only while State = In World
- Highlight Active — Symbolic (True / False); driven by player's Lens Active state

Puzzle Board (Compound Entity)

- Board State — Symbolic (Locked / Open / Complete)
- Pieces Placed — Numeric (0 to total required); triggers win event when maximum is reached
- Unlock Condition — Symbolic (all fragments collected = True)

5.3 Mechanics: Events, Processes, and Conditions

Mechanics describe how entities relate to each other and how the game state changes over time. They are expressed as events (a one-time change triggered by a condition), processes (ongoing sequences that run until stopped), and the conditions that control them.

Relationships Among Entities

- Ghost Alert Level $\pm$ (Player Noise Level $\div$ 10) per second while player is within Detection Radius and ghost Patrol State = Patrolling.
- Ghost Alert Level decays by 5 per second when the player is outside Detection Radius or is crouching out of line of sight.
- Player Noise Level = 10 (crouching), 40 (walking), 80 (running). The relationship between stance and noise is fixed; it does not depend on surface type for the alert calculation, although footstep audio varies by surface material.
- Puzzle Board State changes from Locked to Open when Fragments Collected = total fragment count.

Events

- Fragment Pickup Event: Triggered once when the player presses E within interaction range of a fragment whose State = In World. Fragment State changes to Collected; Fragments Collected increases by 1; the fragment object is removed from the world.
- Ghost Detection Event: Triggered once when any ghost's Alert Level reaches 100. That ghost's Patrol State changes to Alerted; a global difficulty penalty is applied (e.g. all ghost Detection Radii increase by 10%); the event resets when Alert Level drops to 0.
- Puzzle Complete Event: Triggered once when Pieces Placed = total required. Puzzle Board State changes to Complete; a narrative reveal sequence fires; the win state is activated and the game ends.
- Door Interaction Event: Triggered once by a left mouse click when the player's crosshair is on a door within range. The door's State toggles between Open and Closed.

Processes

- Ghost Patrol Process: Initiates at game start. The ghost moves between waypoints in sequence, looping indefinitely. Continues until interrupted by the Ghost Detection Event. Resumes patrol from current position once Alert Level returns to 0.
- Alert Level Accumulation Process: Initiates whenever the player enters any ghost's Detection Radius. Runs continuously, adding to Alert Level each second based on Player Noise Level. Stops when the player exits the radius or crouches out of line of sight.
- Footstep Audio Process: Initiates when the player begins moving. Plays looping footstep audio whose clip is selected based on the surface material directly under the player (wood, stone, or carpet). Stops when the player stops moving.

Conditions

- If (player distance to ghost < Detection Radius) AND (player not crouching out of line of sight) then start Alert Level Accumulation Process.
- Whenever (Alert Level = 100) trigger Ghost Detection Event for that ghost.
- If (Fragments Collected = total fragment count) then unlock Puzzle Board (Board State: Locked $\rightarrow$ Open).
- Whenever (Pieces Placed = total required) trigger Puzzle Complete Event.
- Continue Alert Level Accumulation Process until (player exits Detection Radius) OR (player crouches out of ghost line of sight).

5.4 Internal Economy

The internal economy governs how resources are produced, consumed, and exchanged. Phantom Detective's economy is structured around five mechanisms:

- **Sources** — Memory fragments are placed in fixed world positions at game start, creating a finite, non-replenishing supply. Ghost Alert Level is generated by proximity and noise.
- **Drains** — Ghost Alert Level decays naturally over time. Fragments are permanently removed from the world on collection (converted into inventory).
- **Converters** — Collected fragments are converted into Puzzle Board pieces when placed. Alert Level, once accumulated to 100, is converted into a difficulty penalty (permanent radius increase on all ghosts).
- **Traders** — The player exchanges proximity risk (entering ghost detection zones) for the opportunity to reach fragments. Higher risk yields higher reward potential.
- **Production Mechanism** — The overall loop drives escalating tension: collecting more fragments empties easy rooms first, pushing the player into higher-risk areas, which naturally increases ghost contact and difficulty over time.

5.5 Gameplay from Mechanics: Challenges and Actions

The core mechanics generate active challenges for the player — challenges that require real decisions rather than passive observation. Each mechanic listed above corresponds directly to a player action, and each action either initiates an event or starts or stops a process:

- **Crouching (Ctrl / C)** — Stops the Alert Level Accumulation Process if the player is out of line of sight; also starts the Footstep Audio Process with the quietest clip variant.
 - **Collecting a Fragment (E)** — Triggers the Fragment Pickup Event; increments Fragments Collected; may trigger the Puzzle Board unlock condition.
 - **Toggling the Lens (L)** — Changes Lens Active state; activates or deactivates Highlight Active on all in-world fragments simultaneously.
 - **Opening a Door (Left Click)** — Triggers the Door Interaction Event; toggles the door's state and physically animates the door object.
 - **Placing a Puzzle Piece** — Increments Pieces Placed; if condition is met, triggers the Puzzle Complete Event and activates the win state.
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